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LIFE HISTORY TRAITS IN A POPULATION OF *Pelobates syriacus* (BOETTGER, 1889) FROM TURKEY

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Age structure of the eastern spadefoot toad, *Pelobates syriacus* from the Kızılırmak Delta (Turkey) were assessed using phalangeal skeletochronology. Snout-vent length (SVL) ranged from 42.05 to 86.63 mm in males and 34.03 to 53.27 mm in females. Age of adults ranged from 2 to 8 years in males and 3 to 5 years in females. For both sexes, SVL was significantly correlated with age. Males and females of the toads reached maturity at 2 years of age.

Keywords: longevity; SVL; age at maturity; SSD; Kızılırmak Delta.

INTRODUCTION

Many ecological studies have been performed to learn more about the lifespan of amphibian species (Guarino et al., 2011; Kalaycı et al., 2015; Altunışık, 2018a; Erişmiş, 2018). Investigation of age structure of populations with use of skeletochronology is very important to study lifespan of various species (Li et al., 2017; López et al., 2017; Altunışık, 2018b; Čavlović et al., 2018). In addition, skeletochronological studies were extensively used to compare age structure of different populations of the same species inhabiting different environments (*Rana temporaria*: Guarino et al., 2008; Patrelle et al., 2012; *Rana italica*: Guarino et al., 1995; Buono et al., 2014).

A distributional range of the eastern spadefoot toad, *Pelobates syriacus* (Boettger, 1889) covers the southeastern Balkans (Džukić et al., 2005, 2008), the eastern Caucasus, northern Iran, Turkey, and the Levant (Agasyan et al., 2009; Székely et al., 2013). Individuals of *P. syriacus* are usually active from February until November (Speybroeck et al., 2016).

Pelobates syriacus uses temporary water bodies or large permanent pools for spawning while terrestrial habitats of the species occur in open cultivated landscapes such as light forests, steppes, semi-desert and rocky ar-

eas. Populations of the species are threatened with habitat loss, and they are likely to disappear, especially in arid regions (Agasyan et al., 2009).

In the present paper, we observed body size, age at sexual maturity, and longevity in a breeding population of *P. syriacus* from the Kızılırmak Delta, Turkey.

MATERIAL AND METHODS

Sampling

A population examined is located in the Kızılırmak Delta (41°31.026' N 36°07.397' E), Samsun Province, the Black Sea Region of Turkey. The population is placed in lowland area at altitude of 73 m a.s.l.

Twenty-five adults (12 males and 13 females) were caught by hand on September 6, 2017 (capture permission No. 72784983-488.04-155663 issued by the Ministry of Agriculture and Forestry of Turkish Republic). Specimens were sexed by secondary sex characteristics. Males have a large gland at the back of his fore legs which becomes enlarged in breeding season. All toads were measured from the tip of snout to cloaca (SVL) with a digital caliper with 0.01 mm precision.

The second phalanx of the fourth (longest) toe of a hind limb was clipped in all specimens and preserved in 10% formalin solution for subsequent histologic analyses. After on-site registration and toe-clipping, the

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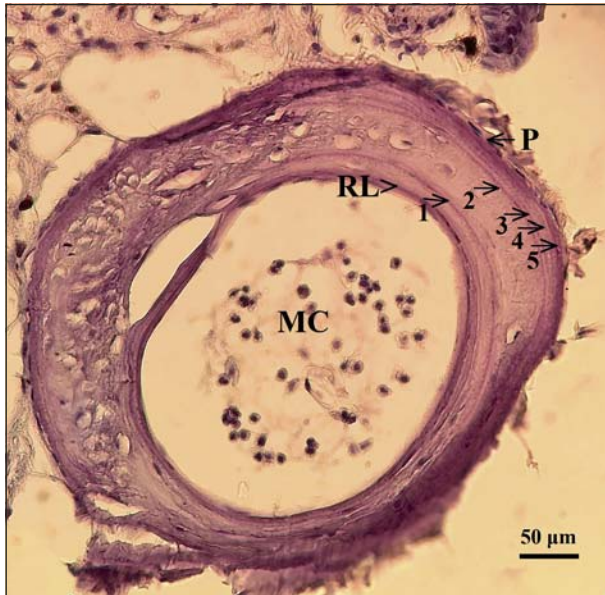


Fig. 1. Cross section (10 μ m thick) of a toe bone of a five-year-old male (62.2 mm SVL) of *Pelobates syriacus* from the Kızılırmak Delta population in Turkey. The age was derived from the presence of five lines of arrested growth (LAGs 1 – 5) surrounding the resorption line: MC, marrow cavity; RL, resorption line; P, periphery.

toads were released back to their natural habitats. The animals were treated in accordance with guidelines of the ethics committee of the Karadeniz Technical University (KTU.53488718-336/2017/21).

Skeletochronology

Standard skeletochronological technique (Castanet and Smirina, 1990; Smirina, 1994) with some minor modifications was used to determine individual age of animals. Phalangeal bones were subjected to decalcification, histological tissue processing, embedding, cross-section, and staining (hematoxylin) procedures as described by Bülbül et al. (2016). Finally, the cross-sections were observed under a light microscope. Age was estimated using standard skeletochronological criteria (Sinsch, 2015; Sinsch and Dehling, 2017).

The observed double lines were counted as single lines as described by Guarino and Erişmiş (2008) and Eroğlu et al. (2018). We diagnosed endosteal resorption of the first LAG by comparing diameters of eroded marrow cavities with diameters of non-eroded marrow cavities in sections from youngest specimens as previously described in the study of Özdemir et al. (2012). In the species studied, age estimation was not seriously affected by the endosteal resorption process. The distance between two adjoining LAGs is a marker of individual

growth in a given year (Kleinenberg and Smirina, 1969; Özdemir et al., 2012). Any apparent decrease in space between two subsequent LAGs was taken as a determiner for age of maturation (Ryser, 1998; Yılmaz et al., 2005; Özdemir et al., 2012). LAG counting was independently performed by two observers. In case of incompatibility the bone section was reexamined by a third observer until consensus was reached.

Statistical Analysis

Since the snout-vent-length (SVL) was normally distributed (one-sample Kolmogorov – Smirnov Test; $P > 0.05$), we used the parametric *T*-test for comparison of means of independent samples. Correlations were estimated with use of the Spearman's rank correlation coefficient (r). Because age classes were not normally distributed for females, the non-parametric Mann – Whitney *U*-test was used for comparison of means. All statistical tests were processed with IBM SPSS 21.0 for Windows at a significance level of $\alpha < 0.05$.

Sexual size dimorphism (SSD) was quantified using the size dimorphism index (SDI) in Lovich and Gibbons (1992). $SDI = (\text{mean length of the larger sex} / \text{mean length of the smaller sex}) \pm 1$. This formula arbitrarily uses +1 if males are larger than females defining the result as negative, or –1 if females are larger than males defining the result as positive.

RESULTS

A growth zone and thin hematoxylinophilic outer line corresponding to a winter line of arrested growth were present in cross sections of the phalanges in 100% ($n = 25$) of adult individuals (Fig. 1). In the adults, we found that endosteal resorption incompletely eroded the first (innermost) periosteal LAGs only in one male. However, the resorption zone did not reach the first LAG in all specimens. The resorption zone clearly seemed out of the endosteal bone in all cross-sections of *P. syriacus* and never created difficulty for age determination. We observed double lines in 9 (36%) specimens. The oldest females and males were 5 and 8 years old, respectively (Fig. 2). The age of start of maturity was 2 years for both sexes in 25 (%100) specimens.

The means of SVL, age and growth rate were 51.0 ± 13.4 mm, 4.16 ± 1.72 years, and 3.21 ± 2.85 for all individuals of *P. syriacus*. In males, the means were 60.0 ± 13.7 mm, 5.00 ± 2.08 years, and 1.71 ± 2.78 . In females, the means were 42.6 ± 5.4 mm, 3.38 ± 0.76 years, and 2.3 ± 1.4 (Table 1).

Age ranged from 2 – 8 years in males and 3 – 5 years in females. The mean age of the specimens was significantly different between the sexes (Mann – Whitney *U*-Test; $Z = -2.11$, $P = 0.035$). Intersexual differences in body size (length) were male-biased ($SSD = -0.409$). The mean SVL was significantly different between sexes ($t = 4.25$, $df = 23$, $P = 0.0003$). There was a strong positive correlation between SVL and age for both males ($r = 0.755$, $P = 0.0050$) and females ($r = 0.848$, $P = 0.0003$).

DISCUSSION

In previous studies, in *P. syriacus* the maximum age was reported by Cogălniceanu et al. (2014) as 11 years for males and 12 years for females (Danube Delta, Romania). Otherwise, Rot-Nikčević et al. (2001) reported the maximum age in a population from Serbia (Utrine) and in a group of populations from Macedonia (Kresna) — Bulgaria (Saramzalino, Sekirnik, and Prdejci) as 16 and 12 years for males and 6 and 15 years for females, respectively. According to our study, the maximum lifespan of the species in Turkey was found as 8 years for males and 5 years for females.

The maximum age in other species of the genus *Pelobates* was provided by various researchers. In Portugal population of *P. cultripes*, the longevity was 8 years in males and 7 years in females (Leclair et al., 2005). While it was 7 and 10 years respectively in Moroccan population of *P. varaldii* (Guarino et al., 2011) and 5 years for both sexes in Italian population of *P. fuscus* (Andreone et al., 2004).

The longevity depends on altitude, latitude and other climatic and environmental factors (Berven, 1982; Miaud et al., 1999). Moreover, the patterns of LAG deposition can vary among populations of amphibians which are subjected to different environmental conditions (Olgun et al., 2005), with tendency to increase mean age and longevity in colder environments.

Mean annual temperature and precipitation over the past 90 years in the Kızılırmak Delta were 14.5°C and 59.72 mm, respectively, according to data of 10th Meteoro-

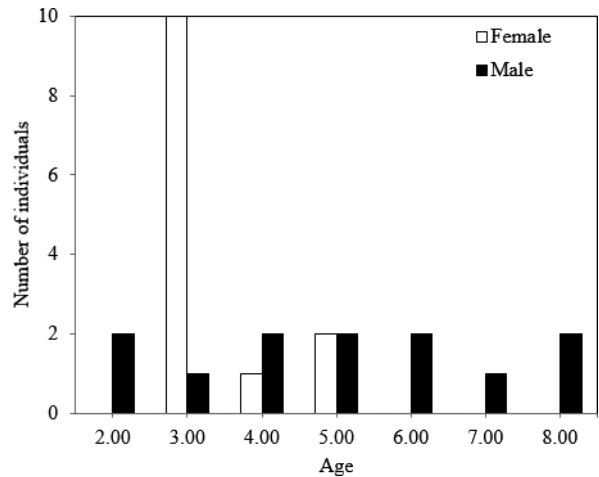


Fig. 2. Frequency distribution of the age in 12 males (black) and 13 females (white) of *Pelobates syriacus* from the Kızılırmak Delta population (Turkey).

logy Regional Directorate, Samsun. During activity period of the toads (from February to November), the mean temperature and precipitation were 16.1°C and 53.35 mm respectively in the study area. Because the altitude, mean annual temperature and precipitation in habitats of the toads in Kızılırmak Delta, Romania and the Balkans (Rot-Nikčević et al., 2001; Cogălniceanu et al., 2014) are nearly similar, the lower lifespan of the toads in the Kızılırmak Delta population could be related to unknown effects, such as predator or anthropogenic press, food availability, etc.

In the present study, the mean age was found as 5.0 years in males and 3.4 years in females. The average age in amphibians is expected to be higher in highland populations (Gül et al., 2011). Although Cogălniceanu et al. (2014) and Rot-Nikčević et al. (2001) also studied lowland populations of *P. syriacus*, they reported higher mean age (7.5 years in males and 7.3 years in females in Romanian population, 6.0 and 4.6 years in Serbian population, and 9.4 and 9.6 years in populations of Macedonia and Bulgaria). These results show that different populations of a same species may have different mean ages. In

TABLE 1. Descriptive Statistics of SVL and Age of the Kızılırmak Delta Population of *Pelobates syriacus*

Characters	Sex	Sample size <i>N</i>	Mean	Range	Standard error SE
SVL, mm	♂♂	12	60.01	41.08 – 86.63	3.94
Age, years		12	5	2 – 8	0.6
SVL	♀♀	13	42.59	34.03 – 53.27	1.50
Age, years		13	3.38	3 – 5	0.21
SVL	♂♂ + ♀♀	25	50.95	34.03 – 86.63	2.67
Age, years		25	4.16	2 – 8	0.34

our study, the mean age was significantly different between males and females. However, Cogălniceanu et al. (2014) did not find differences in the mean age of both sexes. Similar to their findings, Rot-Nikčević et al. (2001) reported differences in the mean age between both sexes only in Utrine population of *P. syriacus* while there were no differences in other populations of the species.

Another important result of the present study was that males of *P. syriacus* in the Kızılırmak Delta population were significantly older than females. Such a trend did not reveal by other researchers (Rot-Nikčević et al., 2001; Cogălniceanu et al., 2014).

As a general rule, among anurans from temperate areas, intraspecific variation in life history traits such as body size, age of maturity and longevity depends from latitude or altitude, which can stem from different climatic influences (Hemelaar, 1988; Sagor et al., 1998; Miaud et al., 1999). Amphibians at high altitudes or high latitudes grow more slowly and take longer to reach sexual maturity than do populations of the same species in warmer climate (Wells, 2007). In our study, it was found that both sexes of the *P. syriacus* reached sexual maturity at 2 years of age. The altitude and latitude of the populations of *P. syriacus* in the study of Rot-Nikčević et al. (2001) are not strongly different from the Kızılırmak Delta population. Accordingly, they reported the age at sexual maturity as 2 – 3 years in both sexes of *P. syriacus* from Serbia, Macedonia and Bulgaria. The age of sexual maturity may vary in different populations of the same species as well as in female and male populations within the same population. Similar to this explanation, Cogălniceanu et al. (2014) reported that age at sexual maturity was 2 – 6 years in males and 2 – 4 years in females of *P. syriacus* from Romania. Age at maturity may also vary among species. However, Leclair et al. (2005), Guarino et al. (2011) and Andreone et al. (2004) reported findings close to our results for age of sexual maturity in other *Pelobates* species (2 – 3 years for both sexes in *P. cultripes*, 2 years for both sexes in *P. varaldii*, and 2 years in males and 3 years females in *P. fuscus*).

We found a strong positive correlation between age and SVL for both females and males of *P. syriacus*. Similar to our findings, Rot-Nikčević et al. (2001) found a significant positive correlation between age and SVL of the species.

In this study, the mean SVL was found higher in males (60.0 mm) than females (42.6 mm) and this difference was significantly different. The SVL may differ between populations from different geographical regions. In anurans, body size is expected to increase in colder areas, being inversely related to climatic factors (i.e., heat and sunlight) directly conditioning individual thermoregulation (Hutchinson and Dupre, 1992). Although there

is not much temperature difference among Turkey and other Balkan countries, Rot-Nikčević et al. (2001) reported higher mean SVL (61.1 mm in males and 60.1 mm in females from Serbian population, 75.8 mm for males and 70.5 mm for females in Macedonian and Bulgarian populations). In addition, Cogălniceanu et al. (2014) also reported higher mean SVL (76.5 mm in males and 74.4 mm in females of the Danube Delta population). The majority of females in the Kızılırmak Delta population consisted of 3 years-old individuals (maximum longevity was 5 years old) and therefore body size of these young females was low. Although in Turkey we found older males (6 – 8 years-old), their number was lower (only 5 male specimens) when number of such old males in studies of Rot-Nikčević et al. (2001) and Cogălniceanu et al. (2014).

SSD in anurans is positively correlated with sex differences in age at maturity (Zhang and Lu, 2013) and sexual differences in growth rates (John-Alder and Cox, 2007; Tomašević-Kolarov et al., 2010). Although we found that age at maturity was similar in both sexes, the male-biased SSD for the Kızılırmak Delta population was statistically significant. The male-biased sexual size dimorphism, which is the unusual state in amphibians, is supposed to be at least partially the result of sexual selection (scramble competition, large-male advantage) in the case of *P. syriacus*. Another important factor on SSD is longevity (Liao and Lu, 2010; Lyapkov et al., 2010). Congruently, we found that longevity was higher in males than females. These results show that longevity may be the main determinant of SSD in the Kızılırmak Delta population of *P. syriacus*.

Double line and endosteal resorption rates might be affected by unpredictable ecological factors (e.g., very high temperatures, very dry conditions, variation in food availability and other occasional environmental changes) (Smirina, 1972; Jakob et al., 2002; Guarino and Erişmiş, 2008). Because the toads in the Kızılırmak Delta population were not exposed to unusually arid conditions (according to data of 10th Meteorology Regional Directorate, Samsun), drought was obviously not the trigger for the double lines observed.

Presence and degree of endosteal resorption in amphibians may vary in populations of the same species inhabiting regions having different weather conditions or daily and annual activity patterns (Esteban et al., 1996). Populations of *Pelophylax perezi* living in warm Mediterranean climate (Hernández and Seva, 1985; Esteban et al., 1996) show a high degree of endosteal resorption. In northern colder regions, no LAGs are destroyed (Patón et al., 1991). Endosteal resorption may also be influenced by altitude. Caetano and Castanet (1993) reported that fewer cases of resorption were observed in lowland pop-

ulations of *Triturus marmoratus* than in populations living at high altitudes. The Kızılırmak Delta population of *P. syriacus* is located in a lowland area (73 m a.s.l.) and the percentage of endosteal resorption was very low (4%). However, high percentages in lowland populations were reported for example in *Pelophylax perezi* and *Pelophylax saharicus* populations (Esteban et al., 1996, 1999). Moreover, endosteal resorption is more marked in aquatic and semi-aquatic frog species than in terrestrial ones (Leclair, 1990; Guarino et al., 1995). Conformably, we found low percentage of endosteal resorption in terrestrial species, such as *P. syriacus*.

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REFERENCES

- Agasyan A., Tuniyev B., Crnobrnja-Isailovic J., Lymberakis P., Andrén C., Cogălniceanu D., Wilkinson J., Ananjeva N., Üzümlü N., Orlov N., Podlousky R., Tuniyev S., and Kaya U. (2009), “*Pelobates syriacus*,” The IUCN Red List of Threatened Species. Version 2015.2. <http://www.iucnredlist.org>, accessed at 2015.08.17.
- Altunışık A. (2018a), “The first demographic data and body size of the southern banded newt, *Ommatotriton vittatus* (Caudata: Salamandridae),” *Acta Herpetol.*, **13**, 13 – 19.
- Altunışık A. (2018b), “Age, Survivorship and Life Expectancy in Near Eastern Fire Salamander, *Salamandra infraimmaculata* (Caudata: Salamandridae),” *Russ. J. Ecol.*, **49**, 166 – 171.
- Andreone F., Bergó P. E., Bovero S., and Gazzaniga E. (2004), “On the edge of extinction? The spadefoot *Pelobates fuscus* insubricus in the Po Plain, and a glimpse at its conservation biology,” *Ital. J. Zool.*, **71**, 61 – 72.
- Berven K. A. (1982), “The genetic basis of altitudinal variation in the wood frog *Rana sylvatica*. I. An experimental analysis of life history traits,” *Evolution*, **36**, 962 – 983.
- Buono V., Guarino F. M., and Vignoli L. (2014), “Maximum body size and age distribution in the Italian stream frog, *Rana italica* Dubois 1987 (Amphibia: Anura),” *Acta Herpetol.*, **9**, 231 – 235.
- Bülbül U., Kurnaz M., Eroğlu A. İ., Koç H., and Kutrup B. (2016), “Age and growth of the red-bellied lizard, *Darevskia parvula*,” *Anim. Biol.*, **66**, 81 – 95.
- Caetano M. H. and Castanet J. (1993), “Variability and microevolutionary patterns in *Triturus marmoratus* from Portugal: age, size, longevity and individual growth,” *Amphibia-Reptilia*, **14**, 117 – 129.
- Castanet J. and Smirina E. M. (1990), “Introduction to the skeletochronological method in amphibians and reptiles,” *Annals Sci. Nat. Zool.*, **11**, 191 – 196.
- Čavlović K., Buj I., Karaica D., Jelić D., and Choleva L. (2018), “Composition and age structure of the *Pelophylax esculentus* complex (Anura; Ranidae) population in inland Croatia,” *Salamandra*, **54**, 11 – 20.
- Cogălniceanu D., Roşioru D., Székely P., Székely D., Buha-ciuc E., Stănescu F., and Miaud C. (2014), “Age and body size in populations of two syntopic spadefoot toads (genus *Pelobates*) at the limit of their ranges,” *J. Herpetol.*, **48**, 537 – 545.
- Džukić G., Beskov V., Sidorovska V., Cogălniceanu D., and Kalezić M. L. (2005), “Historical and contemporary ranges of the spadefoot toads *Pelobates* spp. (Amphibia: Anura) in the Balkan Peninsula,” *Acta Zool. Cracov*, **48A**(1 – 2), 1 – 9.
- Džukić G., Beskov V., Sidorovska V., Cogălniceanu D., and Kalezić M. L. (2008), “Contemporary chorology of the spadefoot toads (*Pelobates* spp.) in the Balkan Peninsula,” *Z. Feldherpetol.*, **15**, 61 – 78.
- Erişmiş U. C. (2018), “Age, size, and growth of the Turkish endemic frog *Pelophylax caralitanus* (Anura: Ranidae),” *Anat. Rec.*, **301**, 1224 – 1234.
- Eroğlu A. İ., Bülbül U., Kurnaz M., and Odabaş Y. (2018), “Age and growth of the common wall lizard, *Podarcis muralis* (Laurenti, 1768),” *Anim. Biol.*, **68**, 147 – 159.
- Esteban M., Garcia-Paris M., and Castanet J. (1996), “Use of bone histology in estimating the age of frogs (*Rana perezi*) from a warm temperate climate area,” *Can. J. Zool.*, **74**, 1914 – 1921.
- Esteban M., Garcia-Paris M., and Castanet J. (1999), “Bone growth and age in *Rana saharica*, a water frog living in a desert environment,” *Ann. Zool. Fenn.*, **36**, 53 – 62.
- Guarino F. M., Angelini F., and Cammarota M. (1995), “A skeletochronological analysis of three syntopic amphibian species from southern Italy,” *Amphibia-Reptilia*, **16**, 297 – 302.
- Guarino F. M. and Erişmiş U. C. (2008), “Age determination and growth by skeletochronology of *Rana holtzi*, an endemic frog from Turkey,” *Ital. J. Zool.*, **75**, 237 – 242.
- Guarino F. M., Gia I. D., and Sindaco R. (2008), “Age structure in a declining population of *Rana temporaria* from northern Italy,” *Acta Zool. Acad. Sci. Hung.*, **54**, 99 – 112.
- Guarino F. M., de Pous P., Crottini A., Mezzasalma M., and Andreone F. (2011), “Age structure and growth in a population of *Pelobates varaldii* (Anura, Pelobatidae) from northwestern Morocco,” *Amphibia-Reptilia*, **32**, 550 – 556.
- Gül S., Özdemir N., Üzümlü N., Olgun K., and Kutrup B. (2011), “Body size and age structure of *Pelophylax ridibundus* populations from two different altitudes in Turkey,” *Amphibia-Reptilia*, **32**, 287 – 292.
- Hemelaar A. S. (1988), “Age, growth and other population characteristics of *Bufo bufo* from different latitudes and altitudes,” *J. Herpetol.*, **22**, 369 – 388.
- Hernández A. and Seva E. (1985), “Datos preliminares sobre la alimentación de la rana común (*Rana perezi*) en la provincia de Alicante,” *Ayudas Investig.*, **3**, 37 – 46.
- Hutchinson V. H. and Dupré R. K. (1992), “Thermoregulation,” in: M. E. Feder and W. W. Burggren (eds.), *Environmental Physiology of the Amphibians*, Chicago Univ. Press, Chicago, pp. 206 – 249.

- Jakob C., Seitz A., Crivelli A. J., and Miaud C. (2002), "Growth cycle of the marbled newt (*Triturus marmoratus*) in the Mediterranean region assessed by skeletochronology," *Amphibia-Reptilia*, **23**, 407 – 418.
- John-Alder H. B. and Cox R. M. (2007), "The development of sexual size dimorphism in *Sceloporus* lizards: testosterone as a bipotential growth regulator," in: D. J. Fairbairn, W. U. Blanckenhorn, and T. Székely (eds.), *Sex, Size and Gender Roles: Evolutionary Studies of Sexual Size Dimorphism*, Oxford Univ. Press, London, pp. 195 – 204.
- Kalaycı T. E., Özdemir N., Altunışık A., and Gül S. (2015), "The effect of altitude, latitude and climatic variables on life-history traits of male *Hyla savignyi* (Audouin, 1827) from Anatolia (Turkey): A skeletochronological study," *Herpetozoa*, **28**(1/2), 55 – 62.
- Kleinenberg S. E. and Smirina E. M. (1969), "On the method of determination of age in amphibians," *Zool. Zh.*, **48**, 1090 – 1094 [in Russian].
- Leclair R. (1990), "Relationships between relative mass of the skeleton, endosteal resorption, habitat and precision of age determination in ranid amphibians," *Annals Sci. Nat. Zool.*, **11**, 205 – 208.
- Leclair M. H., Leclair R. Jr., and Gallant J. (2005), "Application of skeletochronology to a population of *Pelobates cultripes* (Anura: Pelobatidae) from Portugal," *J. Herpetol.*, **39**, 199 – 207.
- Li C., Liao W. B., Yang Z. S., and Zhou C. Q. (2010), "A skeletochronological estimation of age structure in a population of the Guenther's frog, *Hylarana guentheri*, from western China," *Acta Herpetol.*, **5**, 1 – 11.
- Liao W. B. and Lu X. (2010), "Age structure and body size of the Chuanxi tree frog *Hyla annectans chuanxiensis* from 2 different elevations in Sichuan (China)," *Zool. Anz.*, **248**, 255 – 263.
- López J. A., Antoniazzi C. E., Llanes R. E., and Ghirardi R. (2017), "Age structure, growth pattern, sexual maturity, and longevity of *Leptodactylus latrans* (Anura: Leptodactylidae) in temperate wetlands," *Amphibia-Reptilia*, **38**, 371 – 379.
- Lovich J. E. and Gibbons J. W. (1992), "A review of techniques for quantifying sexual size dimorphism," *Growth Dev. Aging*, **56**, 269 – 281.
- Lyapkov S. M., Cherdantsev V. G., and Cherdantseva E. M. (2010), "Geographic variation of sexual dimorphism in the moor frog (*Rana arvalis*) as a result of differences in reproductive strategies," *Zh. Obshch. Biol.*, **71**, 337 – 358 [in Russian].
- Miaud C., Gueyâtant R., and Elmberg J. (1999), "Variations in life-history traits in the common frog *Rana temporaria* (Amphibia: Anura): a literature review and new data from the French Alps," *J. Zool.*, **249**, 61 – 73.
- Olgun K., Uzüm N., Avcı A., and Miaud C. (2005), "Age, size and growth of the southern Crested newt *Triturus karelinii* (Strauch, 1870) in a population from Bozdağ (Western Turkey)," *Amphibia-Reptilia*, **26**, 223 – 230.
- Özdemir N., Altunışık A., Ergül T., Gül S., Tosunoğlu M., Cadeddu G., and Giacoma C. (2012), "Variation in body size and age structure among three Turkish populations of the tree frog *Hyla arborea*," *Amphibia-Reptilia*, **33**, 25 – 35.
- Patón D., Juarranz A., Sequerosm E., Pérez-Campo R., López Torres M., and Barja de Quiroga G. (1991), "Seasonal age and sex structure of *Rana perezi* assessed by skeletochronology," *J. Herpetol.*, **25**, 389 – 394.
- Patrelle C., Hjernerquist M. B., Laurila A., Söderman F., and Merila J. (2012), "Sex differences in age structure, growth rate and body size of common frogs *Rana temporaria* in the subarctic," *Polar Biol.*, **35**, 1505 – 1513.
- Rot-Nikčević I., Sidorovska V., Džukić G., and Kalezić M. L. (2001), "Sexual size dimorphism and life history traits of two European spadefoot toads (*Pelobates fuscus* and *P. syriacus*) in allopatry and sympatry," *Ann. Ser. Hist. Nat.*, **11**, 107 – 120.
- Ryser J. (1998), "Determination of growth and maturation in the common frog, *Rana temporaria*, by skeletochronology," *J. Zool.*, **216**, 673 – 685.
- Sagor E. S., Ouellet M., Barten E., and Green D. M. (1998), "Skeletochronology and geographic variation in age structure in the wood frog, *Rana sylvatica*," *J. Herpetol.*, **32**(4), 469 – 474.
- Sinsch U. (2015), "Review: Skeletochronological assessment of demographic life-history traits in amphibians," *Herpetol. J.*, **25**, 5 – 13.
- Sinsch U. and Dehling L. M. (2017), "Tropical anurans mature early and die young: Evidence from eight Afrotropical *Hyperolius* species and a meta-analysis," *PLoS ONE*, **12**(2), e0171666.
- Smirina É. M. (1972), "Annual layers in bones of *Rana temporaria*," *Zool. Zh.*, **51**, 1529 – 1534 [in Russian].
- Smirina É. M. (1994), "Age determination and longevity in amphibians," *Gerontology*, **40**, 133 – 146 [in Russian].
- Speybroeck J., Beukema W., Bok B., van der Voort J., and Velikov I. (2016), *Field Guide to the Amphibians and Reptiles of Britain and Europe (British Wildlife Field Guides)*, Bloomsbury Natural History, London – New York.
- Székely P., Iosif R., Székely D., Stănescu F., and Cogălniceanu D. (2013), "Range extension for the Eastern spadefoot toad *Pelobates syriacus* (Boettger, 1889)," *Herpetol. Notes*, **6**, 481 – 484.
- Tomašević-Kolarov N., Vljevic L. K., Polovic L. D. G., and Kalezić M. L. (2010), "The body size, age structure and growth pattern of the endemic Balkan Mosor rock lizard (*Dinolacerta mosorensis* Kolombatovich, 1886)," *Acta Zool. Hung.*, **56**, 55 – 71.
- Wells K. D. (2007), *The Ecology and Behavior of Amphibians*, The Univ. of Chicago Press Books, Chicago.
- Yılmaz N., Kutrup B., Çobanoğlu U., and Özorcan Y. (2005), "Age determination and some growth parameters of a *Rana ridibunda* population in Turkey," *Acta Zool. Hung.*, **51**, 67 – 74.
- Zhang L. X. and Lu X. (2013), "Sexual size dimorphism in anurans: ontogenetic determination revealed by an across-species comparison," *Evol. Biol.*, **40**, 84 – 91.